

Bounded slopes and weak access to function corrections

Main theorem and fixed-geometry learning-time consequences

The result below concerns the ordinary sampled tanh readout used in the project's fixed-dictionary formulation [2, Section 8.1]. It bounds the response to an entire space of function corrections, rather than the smallest singular value of a potentially redundant coefficient matrix. The proof specializes classical analytic polynomial approximation [1, Chapter 8]; all constants and the learning-time deductions are given here. No localized comparison network or assumed coefficient witness is needed.

1 Setup and main statement

Let $m, W \geq 1$, $x_i \in [-1, 1]$, and

$$f_v(x) = d + \sum_{j=1}^W c_j \tanh(a_j x + b_j), \quad v = (d, c_1, \dots, c_W), \quad |a_j| \leq \Gamma, \quad \Gamma > 0.$$

All slopes and biases are real. Define the normalized feature matrix

$$A_{i0} = m^{-1/2}, \quad A_{ij} = m^{-1/2} \tanh(a_j x_i + b_j). \quad (1)$$

For $y_i = g(x_i)/\sqrt{m}$, $L(v) = \frac{1}{2} \|Av - y\|_2^2$ is half the empirical mean-squared error. No rank, center-spacing, or target-regularity assumption is imposed.

For an integer $k \geq 0$, set

$$\mathcal{P}_k^X = \{m^{-1/2}(p(x_1), \dots, p(x_m))^T : \deg p \leq k\}, \quad P_k = \text{orthogonal projector onto } \mathcal{P}_k^X. \quad (2)$$

Thus $P_k u = 0$ means that the sample vector u has zero correlation with every sampled polynomial of degree at most k . For $u_i = e(x_i)/\sqrt{m}$, this is equivalent to $m^{-1} \sum_i e(x_i) x_i^\ell = 0$ for $\ell = 0, \dots, k$. The space depends only on the sample locations. If $\mathcal{P}_k^X = \mathbb{R}^m$, its orthogonal complement is trivial and the claims for nonzero orthogonal residuals are vacuous.

Fix a proof parameter $0 < \vartheta < \pi/2$ and define

$$\rho_{\Gamma, \vartheta} = \frac{\vartheta}{\Gamma} + \sqrt{1 + \frac{\vartheta^2}{\Gamma^2}}, \quad M_\vartheta = \max\{1, \tan \vartheta\}, \quad E_k(\Gamma, \vartheta) = \frac{2M_\vartheta}{\rho_{\Gamma, \vartheta} - 1} \rho_{\Gamma, \vartheta}^{-k}. \quad (3)$$

Every choice is valid: $\vartheta = \pi/4$ gives $M_\vartheta = 1$, while choices closer to $\pi/2$ improve the exponential rate at a larger prefactor. This is not a training parameter.

Theorem 1 (Weak access beyond low-degree polynomials). Under the setup above,

$$\left\| A^T (I - P_k) \right\|_2 = \|(I - P_k)A\|_2 \leq \sqrt{W} E_k(\Gamma, \vartheta). \quad (4)$$

Consequently, every $u \in \mathbb{R}^m$ with $P_k u = 0$ satisfies

$$\left\| A^T u \right\|_2^2 \leq W E_k(\Gamma, \vartheta)^2 \|u\|_2^2. \quad (5)$$

Equivalently, every readout perturbation Δv satisfies

$$\|(I - P_k)A\Delta v\|_2 \leq \sqrt{W} E_k(\Gamma, \vartheta) \|\Delta v\|_2. \quad (6)$$

The theorem holds for arbitrary real hidden biases, including centers outside the observation interval, and for rank-deficient or overcomplete dictionaries.

2 Proof of Theorem 1

1. A common bounded analytic extension. Let $\mathcal{E}_\rho$ be the closed region bounded by the Bernstein ellipse

$$z = (\zeta + \zeta^{-1})/2, \quad |\zeta| = \rho > 1.$$

Its imaginary semiaxis is $(\rho - \rho^{-1})/2$. For $\rho = \rho_{\Gamma, \vartheta}$, this equals ϑ/Γ , so every $z \in \mathcal{E}_{\rho_{\Gamma, \vartheta}}$ satisfies

$$|\operatorname{Im}(a_j z + b_j)| \leq \vartheta < \pi/2.$$

The poles of $\tanh w$ lie at $i\pi(\ell + 1/2)$, $\ell \in \mathbb{Z}$. Thus each feature $\phi_j(z) = \tanh(a_j z + b_j)$ is holomorphic on a neighborhood of the closed ellipse. For real s, t ,

$$|\tanh(s + it)|^2 = \frac{\sinh^2 s + \sin^2 t}{\sinh^2 s + \cos^2 t} \leq \max\{1, \tan^2 \vartheta\} = M_\vartheta^2 \quad (|t| \leq \vartheta). \quad (7)$$

This bound is uniform in every real bias b_j .

2. Polynomial approximation of each feature. We include the standard Chebyshev estimate to fix its constant. Suppose F is real-valued on $[-1, 1]$, holomorphic near $\mathcal{E}_\rho$, and bounded there by M . The function

$$H(\zeta) = F((\zeta + \zeta^{-1})/2)$$

is analytic on an annulus containing $\rho^{-1} \leq |\zeta| \leq \rho$. Its Laurent expansion has coefficients $\ell_{-n} = \ell_n$, because $H(\zeta) = H(\zeta^{-1})$. Cauchy's coefficient formula on $|\zeta| = \rho$ yields $|\ell_n| \leq M\rho^{-n}$ for $n \geq 0$. On $\zeta = e^{it}$,

$$F(x) = \ell_0 + 2 \sum_{n \geq 1} \ell_n T_n(x), \quad x = \cos t,$$

where $T_n(\cos t) = \cos(nt)$. The coefficients are real, equivalently by the cosine expansion of $F(\cos t)$. Since $|T_n(x)| \leq 1$, truncating this series at k gives a real polynomial p_k with

$$\|F - p_k\|_\infty \leq 2M \sum_{n=k+1}^{\infty} \rho^{-n} = \frac{2M}{\rho - 1} \rho^{-k}. \quad (8)$$

Apply (8) with (7): each ϕ_j has a degree- k approximant $p_{j,k}$ such that

$$\|\phi_j - p_{j,k}\|_{L^\infty([-1,1])} \leq E_k(\Gamma, \vartheta). \quad (9)$$

3. Project away the polynomial part. Construct a matrix A_k with the *same dimensions and normalization as A* : keep its constant column unchanged and set $(A_k)_{ij} = p_{j,k}(x_i)/\sqrt{m}$ for the hidden columns. Every column lies in $\mathcal{P}_k^X$, so $(I - P_k)A_k = 0$ and $\operatorname{rank}(A_k) \leq k + 1$. It is an auxiliary approximation of A , not another network or a subset of its columns.

The bias-column error is zero, and each hidden-column error obeys

$$\|A_{:j} - (A_k)_{:j}\|_2^2 = m^{-1} \sum_i |\phi_j(x_i) - p_{j,k}(x_i)|^2 \leq E_k(\Gamma, \vartheta)^2.$$

Therefore $\|A - A_k\|_F \leq \sqrt{W} E_k(\Gamma, \vartheta)$, and

$$\|(I - P_k)A\|_2 = \|(I - P_k)(A - A_k)\|_2 \leq \|A - A_k\|_2 \leq \sqrt{W} E_k(\Gamma, \vartheta).$$

Transpose to obtain (4). Applying this operator bound to $u = (I - P_k)u$, or to Δv , proves (5) and (6). $\square$

3 A learning-time lower bound for frozen-geometry training

Corollary 1 (Readout gradient flow). Keep A fixed and train only the raw readout by unit-rate Euclidean gradient flow,

$$\dot{v} = -A^T(Av - y), \quad r(t) = Av(t) - y.$$

Suppose $r_0 = r(0) \neq 0$ and $P_k r_0 = 0$. Then, for every $t \geq 0$,

$$\|r(t)\|_2 \geq \|r_0\|_2 \exp\left[-W E_k(\Gamma, \vartheta)^2 t\right]. \quad (10)$$

For $0 < \varepsilon < 1$, the first time T_ε such that $\|r(T_\varepsilon)\| \leq \varepsilon \|r_0\|$, if it exists, satisfies

$$T_\varepsilon \geq \frac{\log(1/\varepsilon)}{W E_k(\Gamma, \vartheta)^2} = \frac{(\rho_{\Gamma, \vartheta} - 1)^2}{4W M_\vartheta^2} \rho_{\Gamma, \vartheta}^{2k} \log(1/\varepsilon). \quad (11)$$

There is no assumption that $r(t)$ remains polynomial-orthogonal for $t > 0$.

Proof. Let $K = AA^T$. Then $\dot{r} = -Kr$ and $r(t) = e^{-Kt}r_0$. Take an orthonormal eigenbasis q_i of K , with eigenvalues $\lambda_i \geq 0$, and write

$$\hat{r}_0 = r_0 / \|r_0\| = \sum_i \beta_i q_i, \quad \sum_i \beta_i^2 = 1.$$

Convexity of $s \mapsto e^{-2ts}$ and Theorem 1 give

$$\begin{aligned} \frac{\|r(t)\|^2}{\|r_0\|^2} &= \sum_i \beta_i^2 e^{-2t\lambda_i} \geq \exp\left(-2t \sum_i \beta_i^2 \lambda_i\right) \\ &= \exp\left(-2t \|A^T \hat{r}_0\|^2\right) \geq \exp\left(-2t W E_k(\Gamma, \vartheta)^2\right). \end{aligned}$$

Take square roots and rearrange for the hitting time. Zero eigenvalues cause no problem: a component outside the feature span persists, and the required accuracy may never be reached. $\square$

Corollary 2 (Ordinary discrete GD). Under the same initial-residual condition, consider

$$v_{n+1} = v_n - \ell A^T(Av_n - y), \quad 0 < \ell \leq \frac{1}{2\|A\|_2^2}.$$

Then

$$\|r_n\|_2 \geq \|r_0\|_2 e^{-2\ell n W E_k(\Gamma, \vartheta)^2}, \quad N_\varepsilon \geq \frac{\log(1/\varepsilon)}{2\ell W E_k(\Gamma, \vartheta)^2} \quad (12)$$

whenever the relative tolerance ε is reached by step N_ε . The geometry-independent choice $\ell \leq 1/[2(W+1)]$ is sufficient for all dictionaries in Theorem 1.

Proof. The exact residual is $r_n = (I - \ell K)^n r_0$. Every $\ell\lambda_i$ lies in $[0, 1/2]$, where $1 - x \geq e^{-2x}$. Thus

$$\frac{\|r_n\|_2^2}{\|r_0\|_2^2} = \sum_i \beta_i^2 (1 - \ell\lambda_i)^{2n} \geq \sum_i \beta_i^2 e^{-4\ell n \lambda_i} \geq e^{-4\ell n \|A^T \hat{r}_0\|^2} \geq e^{-4\ell n W E_k(\Gamma, \vartheta)^2}.$$

Take square roots. Finally $\|A\|_2^2 \leq \|A\|_F^2 \leq W + 1$, since every normalized column has norm at most one. This proves the common step-size claim. $\square$

4 Meaning, scope, and the measurable quantity

The squared readout response to any unit polynomial-orthogonal correction is bounded by

$$WE_k(\Gamma, \vartheta)^2 = \frac{4WM_\vartheta^2}{(\rho_{\Gamma, \vartheta} - 1)^2} \exp[-2k \operatorname{arsinh}(\vartheta/\Gamma)]. \quad (13)$$

For fixed Γ, ϑ , this decays geometrically with degree. In the large- Γ expansion, the exponent has scale $-2\vartheta k/\Gamma$. The prefactor in (13) remains part of every quantitative claim. For a finite grid, nonzero polynomial-orthogonal probes exist only while $\mathcal{P}_k^X \neq \mathbb{R}^m$.

At any residual $r \neq 0$, frozen-geometry gradient flow satisfies

$$-\frac{d}{dt} \log \|r\| = \frac{\|A^T r\|^2}{\|r\|^2}.$$

For a general residual, rather than assuming exact orthogonality, Theorem 1 gives the directly measurable leakage bound

$$\|A^T r\| \leq \|A\| \|P_k r\| + \sqrt{W} E_k(\Gamma, \vartheta) \|(I - P_k)r\|. \quad (14)$$

It follows simply by splitting $r = P_k r + (I - P_k)r$. Low-degree residual can still drive a substantial gradient. The theorem does not assert that every smooth target or trained residual occupies the weakly accessible subspace.

Paper-level claim. Bounded-slope tanh dictionaries have exponentially weak access to corrections orthogonal to sampled low-degree polynomials. Such initial residuals require the explicit time in (11) or (12) under frozen-geometry readout training. Neither a favorable condition number nor independence of the halo columns is required.

Claim boundary. The operator theorem is unconditional under its slope and domain assumptions; the learning-time corollaries specify the initial residual and freeze the hidden layer. They are exact-arithmetic statements for plain GD in the stated readout coordinates. They do not prove a finite-precision error floor, monotonic improvement with increasing Γ , or failure of jointly trained geometry to escape the bounded-slope region. Evaluating the same polynomial-orthogonal probes across prescribed scales tests the mechanism without assuming a successful localized comparator.

References

- [1] L. N. Trefethen. *Approximation Theory and Approximation Practice*, Extended Edition. SIAM, 2019. Chapter 8, Theorems 8.1–8.2 (Chebyshev coefficients and truncation on a Bernstein ellipse). [doi:10.1137/1.9781611975949](https://doi.org/10.1137/1.9781611975949).
- [2] Continuous MLPs Project. *Quasi-Interpolants as Constructive MLPs on a Finite Interval*, 9 January 2026, Section 8.1. Supplied file `QI_boundary_correction.pdf`. Cited only for the fixed-dictionary readout setup; the structural and dynamical results above are the deductions written out in this note.